

CBSE Test Paper 04
Chapter 6 Work Energy and Power

1. A body of mass 2 kg initially at rest moves under the action of an applied horizontal force of 7 N on a table with coefficient of kinetic friction = 0.1. Compute the (a) work done by the net force on the body in 10 s, (b) change in kinetic energy of the body in 10 s **1**
 - a. 600 J, 600 J
 - b. 555 J, 555 J
 - c. 655 J, 655 J
 - d. 635 J, 635 J
 2. Consider a collision between two pool balls. Ball 1 is at rest and ball 2 is moving towards it with a speed of 2 m s^{-1} . The mass of each ball is 0.3 kg. After the balls collide elastically, ball 2 comes to an immediate stop and ball 1 moves off. What is the final velocity of ball 1? **1**
 - a. 2.2 m/s
 - b. 1.87 m/s
 - c. 2.4 m/s
 - d. 2 m/s
 3. On an essentially frictionless, horizontal ice rink, a skater moving at 3.0 m/s encounters a rough patch that reduces her speed to 1.65 m/s due to a friction force that is 25% of her weight. Use the work–energy theorem to find the length of this rough patch **1**
 - a. 1.3 m
 - b. 1.1 m
 - c. 1.5 m
 - d. 0.9 m
 4. A person trying to lose weight (dieter) lifts a 10 kg mass, one thousand times, to a height of 0.5 m each time. Assume that the potential energy lost each time she lowers the mass is dissipated. Fat supplies $3.8 \times 10^7 \text{ J}$ of energy per kilogram which is converted to mechanical energy with a 20% efficiency rate. How much fat will the dieter use up? **1**
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- a. 8.85×10^{-3} kg
 b. 8.45×10^{-3} kg
 c. 7.45×10^{-3} kg
 d. 6.45×10^{-3} kg
5. A rain drop of mass 2.00 g falls from a height 1.00 km. It hits the ground with a speed of 50.0 m/s. What is the work done by the gravitational force? (Take $g = 10 \text{ m s}^{-2}$) **1**
- a. 26 J
 b. 22 J
 c. 24 J
 d. 20 J
6. Two thermometers are constructed in the same way except that one has a spherical bulb and the other an elongated cylindrical bulb. Which one will response quickly to temperature change? **1**
7. If two bodies stick together after the collision, will the collision be elastic or inelastic? **1**
8. Is it practically possible to have a situation where $(E - U) < 0$? **1**
9. Calculate the difference between two principal specific heats of 1g of helium gas at N. T. P. Given Molecular weight of Helium = 4 and $J = 4.186 \text{ J/cal}$ and Universal Gas constant, $R = 8.314 \text{ J / mole / K}$. **2**
10. Two ball bearings of mass m each, moving in opposite directions with equal speed collide head on with each other. Predict the outcome of the collision, assuming it to be perfectly elastic. **2**
11. A particle of mass 1 kg moving with a velocity $\mathbf{v}_1 = (3\hat{i} - 2\hat{j}) \text{ m/s}$ experiences a perfectly inelastic collision with another particle of mass 2 kg having velocity $\mathbf{v}_2 = (4\hat{j} - 6\hat{k}) \text{ m/s}$. Find the final velocity and speed of the particles. **2**
12. What is a variable force? Find work done by a variable force graphically. **3**
13. The bob of a pendulum is released from a horizontal position. If the length of the pendulum is 1.5 m, what is the speed with which the bob arrives at the lowermost point, given that it dissipated 5% of its initial energy against air resistance? **3**
14. Consider a one-dimensional motion of a particle with total energy E . There are four regions A, B, C and D in which the relation between potential energy V , kinetic energy (K) and total energy E is as given below:
 Region A : $V > E$
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Region B : $V < E$

Region C : $K > E$

Region D : $V > K$

State with reason in each case whether a particle can be found in the given region or not. 3

15. Consider the collision depicted in figure to be between two billiard balls with equal masses $m_1 = m_2$. The first ball is called the cue while the second ball is called the target. The billiard player wants to sink the target ball in a corner pocket, which is at an angle $\theta_2 = 37^\circ$. Assume that the collision is elastic and that friction and rotational motion are not important. Obtain θ_1 . 5

